

IEC 61850 Protocol Reference Manual

1. Absolute Beginner Primer & Protocol Architecture

In legacy substations and solar plants, devices used thousands of hardwired copper cables and proprietary serial protocols like Modbus or DNP3. In Modbus, data lives at plain register numbers (e.g., Holding Register 40001), requiring manual Excel lookup tables.

IEC 61850 replaces copper hardwiring with high-speed Ethernet and introduces an object-oriented framework where variables carry standard self-describing English names (e.g., Breaker1/XCBR1.Pos.stVal), complete with embedded quality flags and nanosecond-accurate UTC timestamps.

Beginner Analogy: Modbus vs. IEC 61850

- **Modbus (Old Way):** Receiving a blank slip of paper with the number 240.5. Is it Voltage? Current? Temperature? You must manually check a spreadsheet.
- **IEC 61850 (New Way):** Receiving a labelled package: Main_Relay/MMXU1.PhV.phsA = 240.5 Volts (Quality: GOOD, Time: 10:00:00.100). Human engineers and SCADA computers read it instantly without manual mapping.

The Three-Level Substation Functional Architecture

Level	Real-World Equipment	Primary Function	Protocol Mapping
Station Level	SCADA Gateway, Local HMI, Engineering PCs	Overall monitoring, historical trending, disturbance log retrieval, remote supervisory commands.	MMS over TCP/IP (Port 102)
Bay Level	Protection Relays, Bay Controllers, Meters	Protection logic (Overcurrent, Distance), interlocking, auto-reclosing, local control decisions.	GOOSE & MMS
Process Level	Merging Units (MU), Breaker Controllers, CTs/VTs	Digitizing raw analog current and voltage waveforms; executing physical breaker trip commands at trip coils.	SV & GOOSE

2. Master Table of Abbreviations & Acronyms

Below is a comprehensive lookup table of all protocol acronyms, data types, and file extensions used in IEC 61850:

Abbreviation	Full Term / Meaning	Abbreviation	Full Term / Meaning
ACSI	Abstract Communication Service Interface	IED	Intelligent Electronic Device
APDU	Application Protocol Data Unit	LD / LN	Logical Device / Logical Node
APPID	Application Identifier	MAC	Media Access Control (Physical Address)
ASN.1	Abstract Syntax Notation One	MMS	Manufacturing Message Specification
BER	Basic Encoding Rules	MU	Merging Unit (Process Level Digitizer)
BESS	Battery Energy Storage System	PDU	Protocol Data Unit
BRCB	Buffered Report Control Block	PPC	Power Plant Controller

Abbreviation	Full Term / Meaning	Abbreviation	Full Term / Meaning
CB / CBR	Circuit Breaker	PRP	Parallel Redundancy Protocol
CID	Configured IED Description (XML File)	PTP	Precision Time Protocol (IEEE 1588)
COTP	Connection-Oriented Transport Protocol	SAS	Substation Automation System
CT / VT	Current Transformer / Voltage Transformer	SBO / SBOes	Select-Before-Operate (enhanced security)
DA / DO	Data Attribute / Data Object	SCADA	Supervisory Control and Data Acquisition
DER	Distributed Energy Resources	SCD	Substation Configuration Description
FC / FCDA	Functional Constraint / FC Data Attribute	SCL	Substation Configuration Language
GoCB	GOOSE Control Block	SSD	Substation Specification Description
GOOSE	Generic Object Oriented Substation Events	SV / SAV	Sampled Values (IEC 61850-9-2)
HMI	Human Machine Interface	TPKT	ISO Transport Service on top of TCP
HSR	High-availability Seamless Redundancy	URCB	Unbuffered Report Control Block
ICD	IED Capability Description (XML File)	VLAN	Virtual Local Area Network (IEEE 802.1Q)

3. Object-Oriented Data Modeling & Logical Node Taxonomy

IEC 61850 organizes data inside an IED into a strict 5-tier object hierarchy:

Object Hierarchy Taxonomy

Physical Device (IED) → Logical Device (LD) → Logical Node (LN) → Data Object (DO) → Data Attribute (DA)

Hierarchy Level	Example Name Component	Technical Role & Description
Physical Device	Relay_Bay01	Physical chassis connected to Ethernet with an IP/MAC address.
Logical Device (LD)	Relay_Bay01/PROT	Functional software domain inside the IED (e.g., Protection LD, Control LD, Measurement LD).
Logical Node (LN)	PTOC1	Standard 4-character functional block. P = Protection, TOC = Time Overcurrent.
Data Object (DO)	Op (Operate) / Pos (Position)	Structured variable representing a specific power grid concept inside the LN.
Data Attribute (DA)	general / stVal / q / t	Primitive leaf containing actual value, quality bitmask, or UTC timestamp.

Example Full Path: Relay_Bay01/PROT.PTOC1.Op.general = Boolean flag showing if Time Overcurrent Protection #1 has operated (tripped).

Standard Logical Node Decoder (Cheat Sheet)

Prefix	Category	Example LNs	Prefix	Category	Example LNs
P	Protection	PTOC, PDIS, PDIFF	X	Switchgear	XCBR (Breaker), XSWI (Isolator)
C	Control	CSWI, CILO (Interlocking)	T	Transformers	TVTR (Voltage), TCTR (Current)
M	Measurement	MMXU (AC), MMDC (Solar DC)	Z	DER / Solar	ZINV (Inverter), ZBAT (Battery)

4. Complete Frame Formats & Protocol Profiles

IEC 61850 uses a hybrid dual-stack approach over the same physical Ethernet cable:

A. GOOSE Frame Structure (Layer 2 - Direct Ethernet Multicast)

GOOSE bypasses TCP/IP layers to deliver critical peer-to-peer trip signals in **under 3 ms**.

ETHERNET II FRAME (802.1Q)			
Destination MAC	Source MAC	802.1Q VLAN Tag	EtherType
01-0C-CD-01-xx-xx (Multicast)	(6 Bytes)	TPID: 0x8100 Priority: 4..7	0x88B8 (GOOSE)
APPID (2 Bytes)	Length (2 Bytes)	Reserved 1 (2 B)	Reserved 2 (2 B)
APDU PAYLOAD (ASN.1 BER)			
- gocbRef: "Bay1_Relay/LLN0\$G0\$ControlGOOSE"			
- timeAllowedToLive: 4000 ms			
- datSet: "Bay1_Relay/LLN0\$DS_Trips"			
- goID: "GOOSE_TRIP_BUS_A"			
- t: Timestamp (UTC with Nanosecond fraction)			
- stNum: 104 (Increments ONLY on status change event)			
- sqNum: 0 (Increments during keep-alive retransmissions)			
- simulation: FALSE			
- confRev: 1			
- ndsCom: FALSE			
- numDatSetEntries: 2			
- allData: [Data Item 1: BOOLEAN = TRUE, Data Item 2: Quality = GOOD]			

B. Sampled Values (SV) Frame Structure (Layer 2 - Direct Ethernet Stream)

SV streams raw digitized voltage and current waveforms from Merging Units to relays (4000 samples/sec at 50Hz).

ETHERNET II FRAME (802.1Q)			
Destination MAC	Source MAC	802.1Q VLAN Tag	EtherType
01-0C-CD-04-xx-xx	(6 Bytes)	Priority: 6	0x88BA (Sampled Val)
APPID (2 Bytes)	Length (2 Bytes)	Reserved 1 & 2 (4 B)	SAV PDU
SV APDU PAYLOAD (ASN.1 BER)			
- noASDU: 1			
- seqASDU:			
- svID: "MU_BAY01_LINE"			
- smpCnt: 1842 (Sample counter 0..3999, resets every second)			
- confRev: 1			
- smpSynch: 1 (Synchronized to IEEE 1588 PTP Master Clock)			
- seqData:			
Ia: Instantaneous Value (INT32) Quality Flags (BITSTRING 32)			
Ib: Instantaneous Value (INT32) Quality Flags (BITSTRING 32)			
Ic: Instantaneous Value (INT32) Quality Flags (BITSTRING 32)			
Va: Instantaneous Value (INT32) Quality Flags (BITSTRING 32)			

C. MMS Frame Structure (Full OSI Stack over TCP/IP)

MMS (ISO 9506) handles client/server communication between SCADA and IEDs over **TCP Port 102**.

--

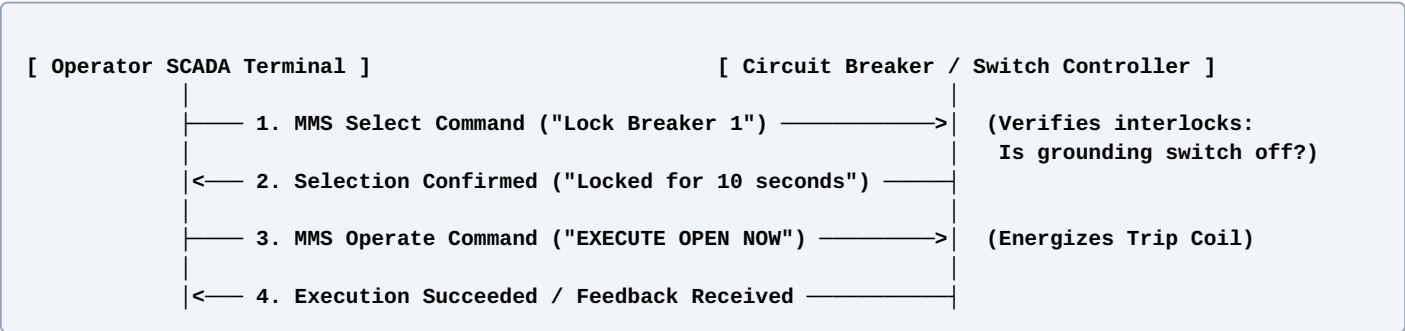
Eth Header	IP Header (Dst IP)	TCP Header (Port 102)	TPKT Header	COTP Header

MMS PDU (ISO 9506 ASN.1 BER)				
- PDU Tag: Confirmed-Request-PDU (Tag: 0xA0) / Response-PDU (Tag: 0xA1)				
- invokeID: 48291 (Sequence identifier matching Request/Response)				
- Service: Read / Write / GetVariableAccessAttributes				
- VariableAccessSpecification:				
- domainSpecific: "Bay1_RelayPROT/PTOC1\$ST\$Op\$general"				

5. ACSI Services, Commands & Control Sequences (SBO)

ACSI Service Class	MMS Mapping	PDU Tag	Description & Practical Function
GetDataValues	Read	0xA0 / 0xA1	Reads current attribute values (e.g., fetch phase current magnitude).
SetDataValues	Write	0xA0 / 0xA1	Writes configuration parameters or protection thresholds into an IED.
Unbuffered Report	InformationReport	0xA3	Spontaneous event push to connected SCADA client; lost if client disconnects.
Buffered Report	InformationReport	0xA3	Spontaneous event push; queues events in IED memory during network outage.
Select / Operate	Write (Structured)	0xA0	Executes breaker open/close under Select-Before-Operate mode.
GetSCLFile	FileOpen / Read	0xA0	Downloads SCL files or COMTRADE fault disturbance recordings.

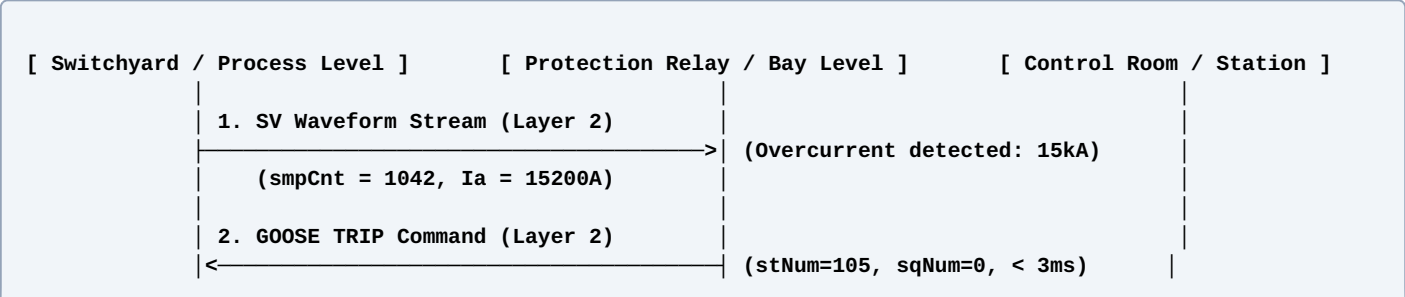
Select-Before-Operate (SBO) Sequence

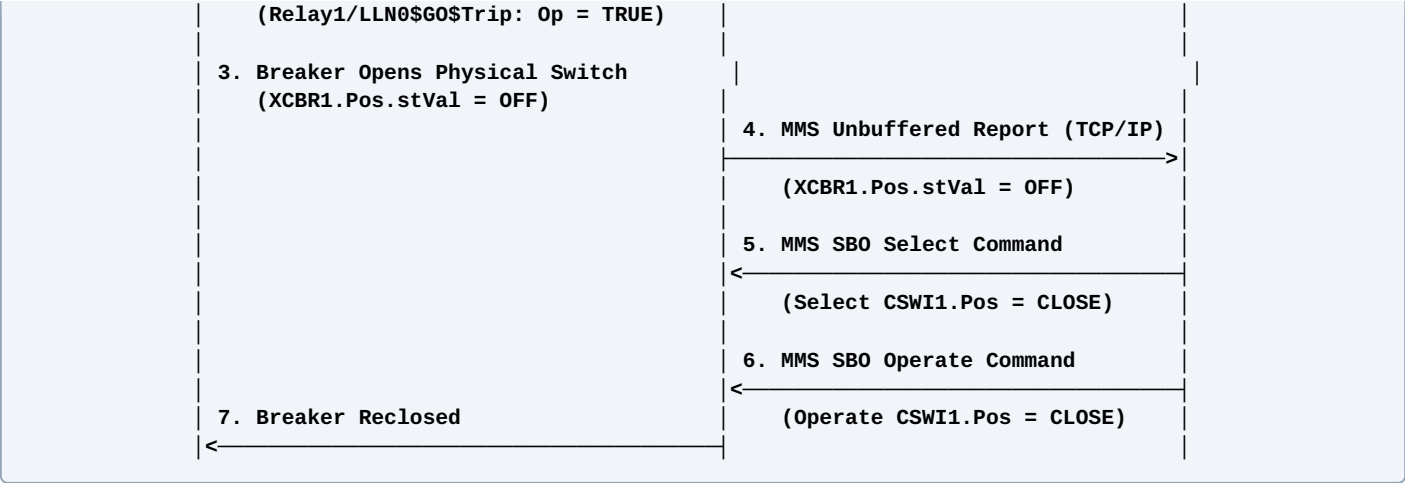


6. End-to-End Operational Example Across All 3 Stacks

Operational Fault Sequence

A heavy short-circuit fault (15 kA) occurs on Transmission Line 1. The Merging Unit reads current, the Protection Relay trips via GOOSE, and SCADA is notified via MMS Report.





7. Physical Media, Redundancy (PRP/HSR) & PTP Time Sync

- **Optical Fiber (Primary Backbone):** Multimode/Singlemode 100Base-FX / 1000Base-LX using ST or LC connectors. Provides complete immunity to Electromagnetic Interference (EMI), lightning surges, and ground potential rise.
- **Shielded Copper (STP):** Cat 6A / Cat 7 STP cables for short panel wiring (< 100 m).

Zero-Loss Redundancy Protocols (IEC 62439-3)

Redundancy Standard	Network Topology	Failover Delay	Operational Mechanism
PRP (Parallel Redundancy)	Dual Independent LANs (LAN A + B)	0 ms (Seamless)	Source duplicates every frame and sends over LAN A & B simultaneously. Receiver accepts first, drops second.
HSR (High-availability Ring)	Ring Topology	0 ms (Seamless)	Nodes forward duplicate frames in clockwise and counter-clockwise directions around the ring.

Time Synchronization: Sampled Values (SV) require sub-microsecond precision (< 1 microsecond). IEC 61850 uses **IEEE 1588v2 Precision Time Protocol (PTP) Power Profile (IEC/IEEE 61850-9-3)** over Ethernet.

8. Detailed Step-by-Step Connection Setup & Configuration

SCL Engineering Files Architecture (IEC 61850-6)

- **.ICD (IED Capability Description):** Vendor template file defining default IED capabilities.
- **.SSD (Substation Specification Description):** Single-line diagram electrical specification.
- **.SCD (Substation Configuration Description):** Master system file containing all devices, IP addresses, and dataset bindings.
- **.CID (Configured IED Description):** Instantiated file downloaded directly into a specific IED.

Step-by-Step Setup Workflow

- Step 1: Managed Switch Setup:** Configure VLAN 10 (GOOSE, Priority 5), VLAN 20 (SV, Priority 6), VLAN 1 (MMS SCADA). Enable IEEE 1588v2 PTP Boundary Clock mode on all switches.
- Step 2: System Configuration (.SCD):** Import .ICD files into System Configurator (e.g., DIGSI, PCM600). Assign static IPs (e.g., Relay 192.168.1.11, SCADA 192.168.1.100). Create GOOSE Dataset DS_TRIP, assign Multicast MAC 01-0C-CD-01-00-01 and APPID 0x0001. Map publisher outputs to subscriber inputs.
- Step 3: Device Loading:** Download generated .CID file into IED via FTP or engineering software. Reboot device and verify status reads RUNNING.

4. **Step 4: SCADA MMS Driver Setup:** Import master .SCD into SCADA system (e.g., Zenon, WinCC). Driver automatically creates tags targeting 192.168.1.11:102. Enable Buffered Report Control Blocks (BRCB).
5. **Step 5: Wireshark Testing:** Connect PC to mirror port. Filter eth . type == 0x88b8. Verify background keep-alive GOOSE frames arrive every 4000 ms. Inject fault current with secondary injection test set (e.g., Omicron) to confirm GOOSE trip bursts arrive in < 3 ms.

9. Solar PV Plant & DER Integration (IEC 61850-7-420)

In utility-scale solar farms, IEC 61850-7-420 extends standard logical nodes to Distributed Energy Resources (DERs):

Solar Component	Logical Node	Protocol	Functional Purpose
Solar Inverter	ZINV / ZCON	MMS	Controls active power limits (W), reactive power (VAR), and Volt-VAr curves.
Solar DC Array	MMDC	MMS	Measures DC voltage and current directly coming from solar panel strings.
AC Measurement	MMXU	MMS / SV	Streams 3-phase AC voltage, current, frequency, and power factor.
Weather Station	MMET	MMS	Measures solar irradiance (W/m²), panel temperature, and wind speed.
Battery (BESS)	ZBAT / ZBTC	MMS / GOOSE	Monitors State of Charge (SoC), State of Health (SoH), and manages charging cycles.
Circuit Breakers	XCBR	GOOSE	Sub-3ms protective tripping during collector feeder short-circuits.